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SYSTEM ARCHITECTURE AND DESIGN PARAMETERS FOR EXTRUSION-BASED AUTONOMOUS CONSTRUCTION SYSTEMS

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ABSTRACT

Autonomous construction systems (ACSs) have become a topic of great interest in recent years in a variety of areas, including design, materials science, architecture, space exploration, natural disaster recover, military operations, and others. Several different approaches have been proposed, the most promising (and so far most widely-applied) one being a large-scale system based on additive manufacturing (or 3-D printing) principles, where a concrete- or foam-based material is extruded in layers to produce a structure. This structure may be used as a basic shell around which a useful building, shelter, bridge, extraterrestrial habitat, or other infrastructure can be built or may be able to produce a full building in one operation. This article extracts information about the the major components, subsystems, and interfaces in these systems from a broad sampling of published literature and uses this information to propose a quasi-general system architecture and identify design opportunities. These models can be used to drive further research efforts on these systems, assist with more agile implementation, and improve the design of large-scale 3-D printing-based systems. This work is a first step in the development of a reliable general system architecture similar to those used in the design of large-scale military and aerospace systems.

1 INTRODUCTION

Autonomous construction systems have become a topic of great interest in recent years in a variety of areas, including design [1], material mechanics [2, 3], building design and modularization [4], sustainability [5], space exploration [6], natural disaster recovery [7], military operations [8], and others. Several different approaches have been proposed, one of the most promising (and so far most widely-applied) one being a large-scale system based on extrusion-based additive manufacturing principles [2, 3, 9–14]. The typical materials used include concrete, asphalt, and ceramic and polymer foams. The resulting structure can be a basic shell around which a useful building, shelter, bridge, extraterrestrial habitat, or other infrastructure can be built, may service as a form for concrete work, or may be able to produce a full building on its own in one operation. In this work, these systems will be referred to as extrusion-based autonomous construction systems (EBACSs). Examples of EBACS components and operations are shown in Figure 1. A number of specific EBACS technologies exist (some examples include classic contour crafting [15], cable-driven contour crafting (six degrees of freedom) [16], Minibuilders [17], Digital Construction Platform (DCP) [18], CONPrint3D [14], and WASP [19]), but they all have many common features and objectives. In order to more effectively design and apply these systems to real-world building problems, it is necessary to understand the commonalities between these system and the important interfaces within them. Therefore, a systems engineering approach should be taken to understand these systems and to develop a high-level general system architecture which can be applied and adjusted as needed for particular applications. Designers of these systems can then use the general system architecture as a starting place for implementation for specific applications. This will not only prove to be a powerful tool for decision making and design when using EBACSs, but will make their implementation far more agile and efficient. The increased quality of decision-making information available to users will reduce cost and schedule risks. Finally, lessons-learned with these systems may be applied to the development and refinement of other autonomous construction systems, such as those which rely on mobile robots and modular building components which can be assembled autonomously [20–26].

The development of a general system architecture for EBACS is a large problem that will take extensive thought and development. However, looking at the existing systems and extracting out their common features, important design parameters, and interfaces and mapping these to generate a quasi-general architecture provides an excellent starting place. As such, this article represents the first steps in developing a general system ar-

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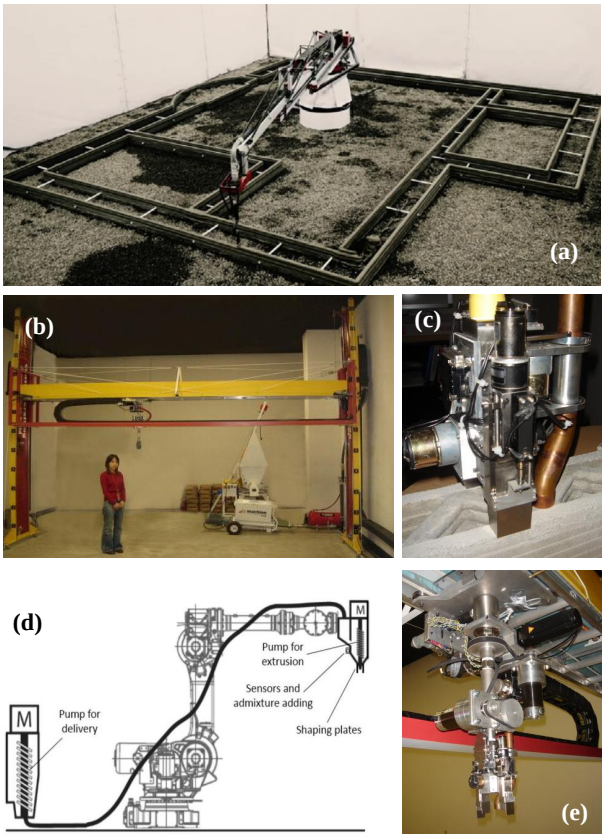

FIGURE 1: *(a-b) Example EBACS, (c) typical extruder, (d) example material mixing and transport system, and (e) typical extruder orientation system. Figures from Ref [27–29] © Elsevier B.V., reproduced with permission.*

chitecture for EBACSs. This study had five major objectives:

1. Identify and collect a representative sample of published papers describing the system architecture of EBACSs.
2. Extract the important features, design parameters, and interfaces within the various existing systems.
3. Map common features, parameters, and interfaces into a quasi-general system architecture (based on the existing systems).
4. Examine and discuss gaps and needed research areas for EBACSs and identify the next steps needed for the development and refinements of a general system architecture for EBACSs.
5. Identify opportunities for improvements in design and implementation for EBACSs.

The first two objectives are examined in Section 3, Objective 3 is discussed in Section 4, and the final two objectives are covered in Section 5. Finally, some final remarks and suggestions for future needed research are given in Section 6.

2 METHODOLOGY

2.1 Research Questions

The study objectives determined the research questions for this work. Specifically, the research questions were: Given the collected set of literature,

1. What are the major functions completed by all the identified EBACSs?
2. What are the major system components or subsystems common to all the identified EBACSs?
3. What are the major interfaces within the system that are present for all the identified systems?
4. Given the collected information, what is the general or quasi-general (since the dataset is based on a sampling of existing systems) architecture for all examined EBACSs?
5. What are the major design parameters for EBACSs that can be tuned and applied directly to specific construction problems?

After answering the research questions via literature review, the results are discussed and insights offered by the authors in Section 5 and Section 6.

TABLE 1: *Study assumptions and their potential risk and impact on the developed design, architecture, and interfaces if wrong*

Assumption	Impact	Risk
The collected materials for the study represent the state-of-the-art for EBACSs	Missed information	Low
No major EBACS types were missed during the review	Missed information	Low
The system architecture, design, and interfaces within the systems were accurately and clearly described in the reviewed papers	Inaccurate understanding of architecture, design, and interfaces	Low
Only a sampling of papers for each type of system was used was enough to understand the operation, construction, and interfaces in the system and so a comprehensive review was not necessary	Missing information about small system variants and developments	Low

2.2 Scope and Limitations

It should be noted that, since this study was meant to collect data about the design and architecture of EBACSs and not to review the full state-of-the-art of the technology, all available papers for a given method were not reviewed. The most obvious example of this was for the contour crafting process; in this case, dozens of papers exist to document small improvements in the technology but only the ones that specifically discussed the system architecture and major components/interfaces were included. This prevented a large amount of redundant or irrelevant information from being included in the study and helped keep it focused on answering the specific research questions.

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TABLE 2: *Papers and references used to extract EBACS design and architecture data.*

Ref	Authors	EBACS Type	Build Type	Material Type
[15]	B. Khoshnevis	Gantry-based contour crafting	Small to medium ceramic shells for concrete pouring	Ceramic paste
[27]	J. Zhang, B. Khoshnevis	Gantry-based contour crafting	Medium to large walls	Concrete
[16]	P. Bosscher, et al.	Cable-driven contour crafting	Medium to large walls	Concrete
[2]	H. Alhumayani, et al.	Robotic arm	Medium to large walls	Fiber-reinforced concrete
[30]	WASP	Delta printer	Small to large scale printing	Ceramic paste, clay, concrete, fiber-reinforced concrete
[2]	H. Alhumayani, et al.	Delta printer	Medium to large walls	Cob/clay
[3]	D. Asprone, et al.	Delta printer	Beams and trusses	Fiber-reinforced concrete
[31]	O. Kontovourkis, G. Tryfonos	Robotic arm	Small to medium sized shells and structures	Clay
[32]	S. Ercan, et al.	Robotic arm	Small to medium structures	Plaster
[33]	J. Burger, et al.	Robotic arm	Small to medium structures	Fiber-reinforced concrete
[34]	APIS-COR	Adjustable boom	Large walls and structures	Concrete
[35]	E. Barnett, C. Gosselin	Cable-suspended foam extrusion	Small to medium structures	PU foam
[36]	B. Furet, et al.	Robotic arm	Large walls	PU foam, concrete
[13]	C. Gosselin, et al.	Robotic arm	Small to medium structures	High-performance concrete
[37]	E. Lublasser, et al.	Robotic arm	Wall coatings and repair	Ceramic foam
[14]	V. Mechtcherine, et al.	CONPrint3D with truck-mounted boom	Very large walls and monolithic structures	Concrete
[18]	S. Keating, et al.	Digital Construction Platform (DCP) with boom-mounted robotic arm	Forms for poured concrete and insulation	PU foam
[38]	S. Keating, N. Oxman	Robotic arm	Small to medium structures and concrete forms	PU foam
[39]	S. Jokic, et al.	Minibuilders: Team of smaller robotic arms	Small to medium structures	Fiber-reinforced concrete
[17]	X. Zhang, et al.	Minibuilders: Team of smaller robotic arms	Small to medium structures	Fiber-reinforced concrete
[40]	R. Wolfs, et al.	Gantry-based contour crafting	Small to medium structures	Concrete
[41]	S. Lim, et al.	Gantry-based contour crafting	Medium to large structures with curved/contoured surfaces	Concrete, ceramic paste
[42]	GreenPrint	Gantry-based contour crafting	Medium to large structures	Concrete, ceramic paste

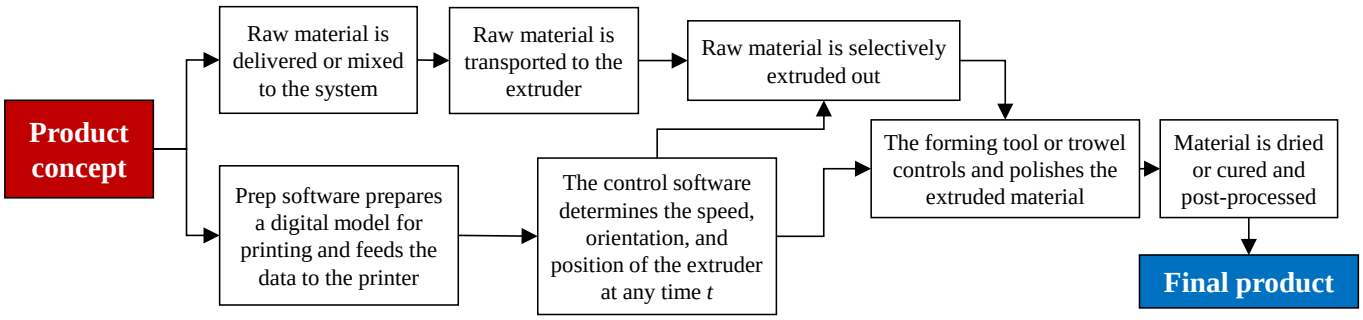

FIGURE 2: *Identified EBACS process flow*

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2.3 Data Collection and Screening

This work relied on the collection of a set of data from published papers on extrusion-based autonomous construction systems (EBACSs). A set of keywords (available upon request from the corresponding author) was established to search for references in Google Scholar and Scopus, followed by a general Google search to gather any missing information. Using these keywords, the initial set of papers was selected based on the relevance and title to the research questions, resulting in a collection of 42 papers. After screening these for relevance to the research questions, the final set of 23 sources was used to collect data for this study. The dataset is discussed in Section 3. During the final screening, papers not in English, conference papers for which an equivalent journal version was available, and papers published in potentially predatory journals were excluded. Papers and sources with heavy commercialization were only included when they were the only available source of useful information.

2.4 Impact of Assumptions

Since the study was based on a dataset collected from a sampling of published papers, several assumptions had to be made. The conclusions of this article should be understood in light of these assumptions and their potential impact if wrong. The assumptions and their potential impact are described in Table 1.

The four major assumptions were determined to present a low risk to this study, given that the collection and screening of the papers was done carefully to avoid missing information. In addition, all of the collected studies for review were high-quality and well-described and so were useful for information extraction.

3 COLLECTED DATA

3.1 Reference Set

The collected references are presented in Table 2. This table shows the 23 final references by EBACS type, build type (including standard print size), and material type (concrete, ceramic paste, polyurethane (PU) foam, and others). This information, combined with the process flow shown in Figure 2, and identified subsystems and interfaces (Figure 3), was used to extract out a general system architecture and set of system design parameters in order to answer the research questions for this study.

3.2 Process Flow

The process flow for this can be seen in Figure 2. These tasks and flow help to establish the subsystems and the interfaces in the system, which are shown in Figure 3.

3.3 Fundamental System Tasks

All extrusion-based autonomous construction systems (EBACSs) reviewed were noted to have five fundamental tasks:

1. Collect raw materials, mix them appropriately for the task at hand. These materials typically consist of various concrete and foam mixtures.
2. Transport these materials to a nozzle or applicator which will extrude the raw material out.
3. Selectively place the raw material to build walls and other geometry.
4. Adjust, tune, or otherwise ensure proper placement of the materials via a guide or trowel.
5. Monitor and control the process from the beginning to completion, with or without direct human supervision.

3.4 Subsystems

Given the process flow and the five major objectives previously discussed a general EBACS can be divided into eight subsystems, two software subsystems and six hardware subsystems. This subsystem breakdown can be seen in Figure 3.

1. **Subsystem 1**: The frame or support used for holding, controlling, and positioning the extruder. In the collected data, this subsystem varied widely from a simple gantry to a robotic arm to a team of small printing robots to a truck-mounted boom.
2. **Subsystem 2**: The preparation software (to model, slice, and prepare the model for printing).
3. **Subsystem 3**: The material mixing and transport system.
4. **Subsystem 4**: The extruder.
5. **Subsystem 5**: The orientation system for the extruder. In all EBACS cases reviewed, the orientation of the extruder was vital to effective and efficient printing. While the extruder itself was mostly the same on all the reviewed systems, the orientation system varied and so should be considered its own subsystem.
6. **Subsystem 6**: The forming or troweling tool, which may be connected to the extruder or independently controlled.
7. **Subsystem 7**: The control software and firmware needed during printing.
8. **Subsystem 8**: The control hardware. This includes sensors, cameras, and other hardware used to monitor and control the other components during printing.

A ninth subsystem could be conceived for the interface between the system and a human operator or monitor. However, this was not discussed for EBACSs in any depth in the reviewed papers and so is not included here. This will be a major point of discussion in Section 5 and Section 6.

3.5 System Interfaces

Upon reviewing the papers and other references outlined in Table 2, it was noted that the interfaces within any of the EBACSs could be broken down into one of three types:

1. **Mechanical interfaces**: Direct hardware interfaces within the system.
2. **Data/information interfaces**: Data and information communication with the system and between subsystems.
3. **Control interfaces**: Hardware and software interfaces that directly controlled things such as the position, print pattern, material extrusion rate, and other print parameters, as well as monitoring the progress of the build and looking for defects in some cases. Note that in this analysis, hardware which

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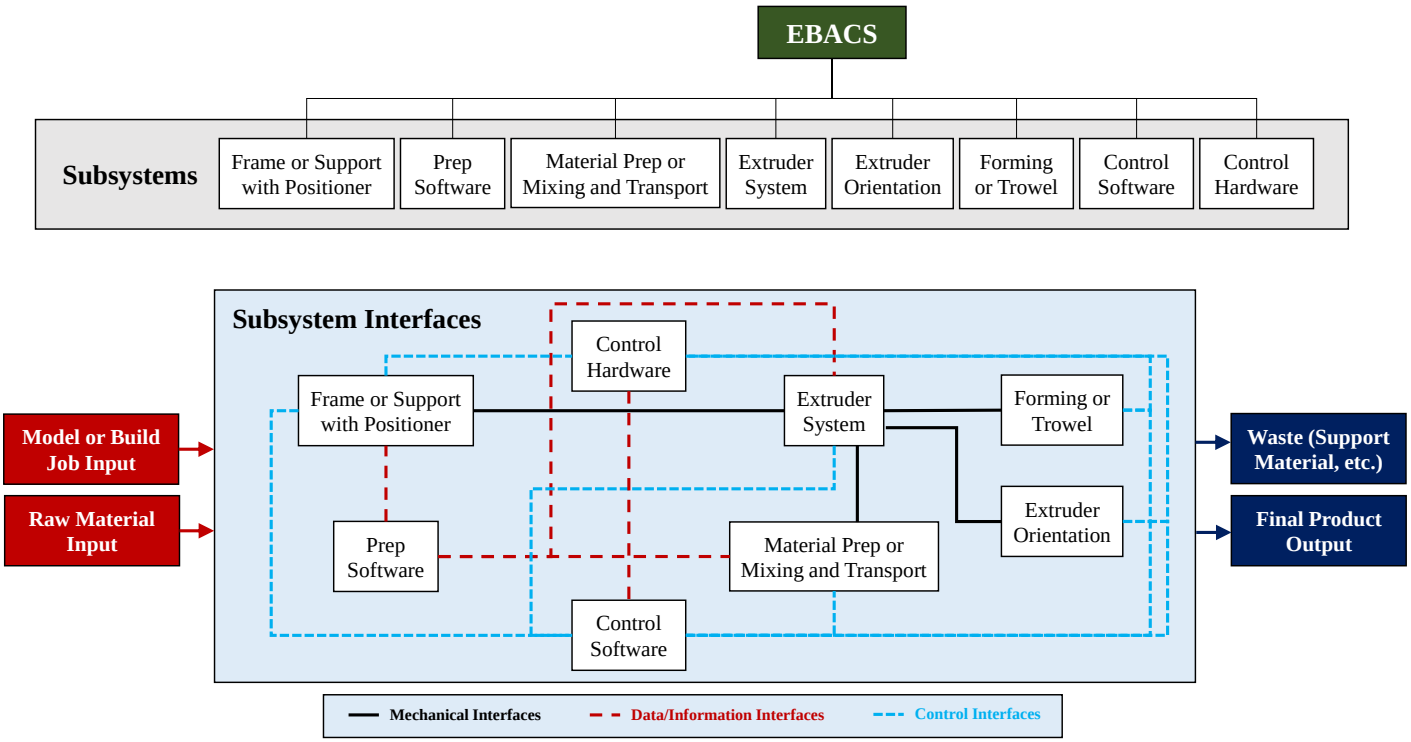

FIGURE 3: *Identified EBACS subsystems and interfaces between these subsystems.*

existed in the system soley for the purpose of control (such as sensors) was included under control interfaces and not mechanical interfaces. These could be validly considered to be mechanical interfaces and this designation is simply the choice of the authors.

3.6 Fundamental Components

Each of the subsystems should be considered to be fundamental components, as they are all necessary for operation and were clearly present in all the reviewed systems shown in Table 2.

3.6.1 Frame and Positioning System Several different frame and positioning systems have been successfully used within the reviewed works. These can be classified into six different categories, as shown in Table 3. Four fundamental components (either by direct observation or knowledge of positioning systems by the authors) were identified as being necessary for all EBACSs. These include (1) the frame itself, (2) the positioners (belts, lead screws, gearbox, etc.) for each axis of motion, (3) the driver motors (stepper motors, servos, and similar), and (4) the power supply or other power input. Other hardware could be present and would be dependent on the exact design of the system at hand.

3.6.2 Prep Software Table 4 shows the software package used (if specified in the paper) for each of the reviewed systems. In most cases, a custom or proprietary software was used or the package used was not specified in the reviewed papers.

TABLE 3: *Frame and positioning systems on existing EBACSs.*

Frame and Positioning System	References
XYZ Gantry	[15, 27, 40–42]
Cable-driven system	[16, 35]
Delta robot/printer	[2, 3, 30]
6 DoF robotic arm	[2, 13, 18, 31–33, 36–38]
Fixed or mounted boom	[14, 34]
Team of small printing robots	[17, 39]

As with other extrusion-based additive manufacturing processes, four distinct preparation software functions are required: (1) A modeling software to represent the final product, (2) a technique for converting the model into a printable file, (3) a method for slicing the model, and (4) toolpath or g-code generation to determine the build process.

TABLE 4: *Prep software packages used for existing EBACSs.*

Prep Software	References
Ultimaker Cura	[14]
COMPASS FAB	[32]
In-house proprietary software	[2, 3, 17, 18, 27, 30, 31, 33–35, 38–42]
Not clearly specified	[13, 15, 16, 36, 37]

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3.6.3 Material Prep and Transport

In all cases where the material preparation method was specified, the material was premixed and fed into the printer system already prepared. The transport system was generally either not discussed in depth or consisted of a standard concrete or foam pump. No special or novel technology in this area was noted when reviewing the collected studies.

3.6.4 Extruder System

While several different extruder designs exist in the reviewed papers, all of them have the same general functionality and most of the differences are related to the shape of the extruded material element and the speed of printing. In the references where the extruder system was described, five basic parts were shown (Figure 1c and 1e); these are very similar to those required for other extrusion-based additive manufacturing processes and so were familiar to the authors. These include (1) the extruder mount (the interface between the positioner and extruder, (2) the extruder assembly itself, (3) the interface with the material handling system through which the raw material is supplied to the extruder, (4) the holding chamber for the raw material before being sent to the extruder, (5) and the extruder nozzle itself, which drives many of the print parameters. Other components may be present depending on the specific design and layout of a particular EBACS.

3.6.5 Extruder Orientation

The extruder orientation was observed to be vital to the effective operation and placement of the extruded materials. In the cases where the positioning was done using a robotic arm, one of the robot head's rotational degrees of freedom was dedicated to this purpose. In all other cases, the extruder orientation was controlled either by a separate motor or a cable/wire.

3.6.6 Forming or Trowel

When reviewing the collected references, it was expected that the forming/trowel system would be prominently discussed, due to its vital role in controlling and forming the final print. However, very little detail was given; the cases where it was discussed, the system was mounted to the end of the extruder head. It was not clearly specified in most of these cases whether it was independently controlled during processing or if it was also controlled as part of the extruder orientation. A lack of details about the trowel design and function was a major weakness observed in the reviewed papers and should certainly be improved in future works.

3.6.7 Control Software

Table 5 shows the control software package used (if specified in the paper) for each of the reviewed systems. The control software includes several essential components and was observed to be very similar to classic extrusion-based additive manufacturing. These include the (1) positioner firmware, (2) the extruder firmware, (3) the generated toolpath, (4) any kind of human-system interface, and (5) any kind of feedback, real-time control, and correction systems (including vision systems if relevant).

TABLE 5: *Control software packages used for existing EBACSs.*

Control Software	References
MATLAB/Simulink	[35]
Open-source robot control software	[17, 18, 31, 32, 38, 39]
In-house proprietary software	[14, 15, 27, 30, 33, 34, 40–42]
Not clearly specified	[2, 3, 13, 36, 37]

3.6.8 Control Hardware

Control hardware was another area where the authors expected to find a significant amount of information in the reviewed studies. However, beyond a couple of the references [31, 35], very little was discussed and so the control hardware systems are not well-known. This can be partially explained by the fact that many of the studies used existing proprietary systems or used systems based on purchased robotic arms and so details of the control hardware were either not known or not considered relevant to the study by the reviewed authors. In addition to the forming/trowel design, the design of control hardware is another major area that needs to be developed and refined for a final general system architecture for EBACSs.

3.7 System Design Parameters

Extusion-based autonomous construction systems (EBACSs) have a large number of system design parameters. This is clear from reviewing the essential components, interfaces, and control systems discussed in previous sections. In fact, the most important design parameters largely correspond with the essential system components, interfaces, and control systems. This will be dicussed further in Section 5.

4 EBACS SYSTEM ARCHITECTURE

Taking into consideration the process flow, system interfaces, the subsystems, the essential components, and other architecture information from the reviewed studies, a quasi-general system architecture for EBACSs can be generated. This is not a general system architecture due to uncertainties in some areas (particularly control hardware and forming/troweling operation) and the fact that it was extracted from a sampling of papers on the topic. However, it is a good first step in developing a general system architecture and is very likely to be highly representative of the final system architecture due to the care taken and the deep analysis undertaken during this study. From here, other sources of information will need to be used to further refine the architecture. This quasi-general system architecture is shown in Figure 4.

5 ANALYSIS AND DISCUSSION

The purpose of this study was to work toward a general system architecture and clear set of system design variables for extrusion-based autonomous construction systems (EBACSs) by examining a wide sampling of the existing literature. This study explored the common features, patterns, and available technical information in order to extract out a quasi-general architecture that was common to and applied to the wide variety of systems studied (Table 2). Five research questions were defined at the

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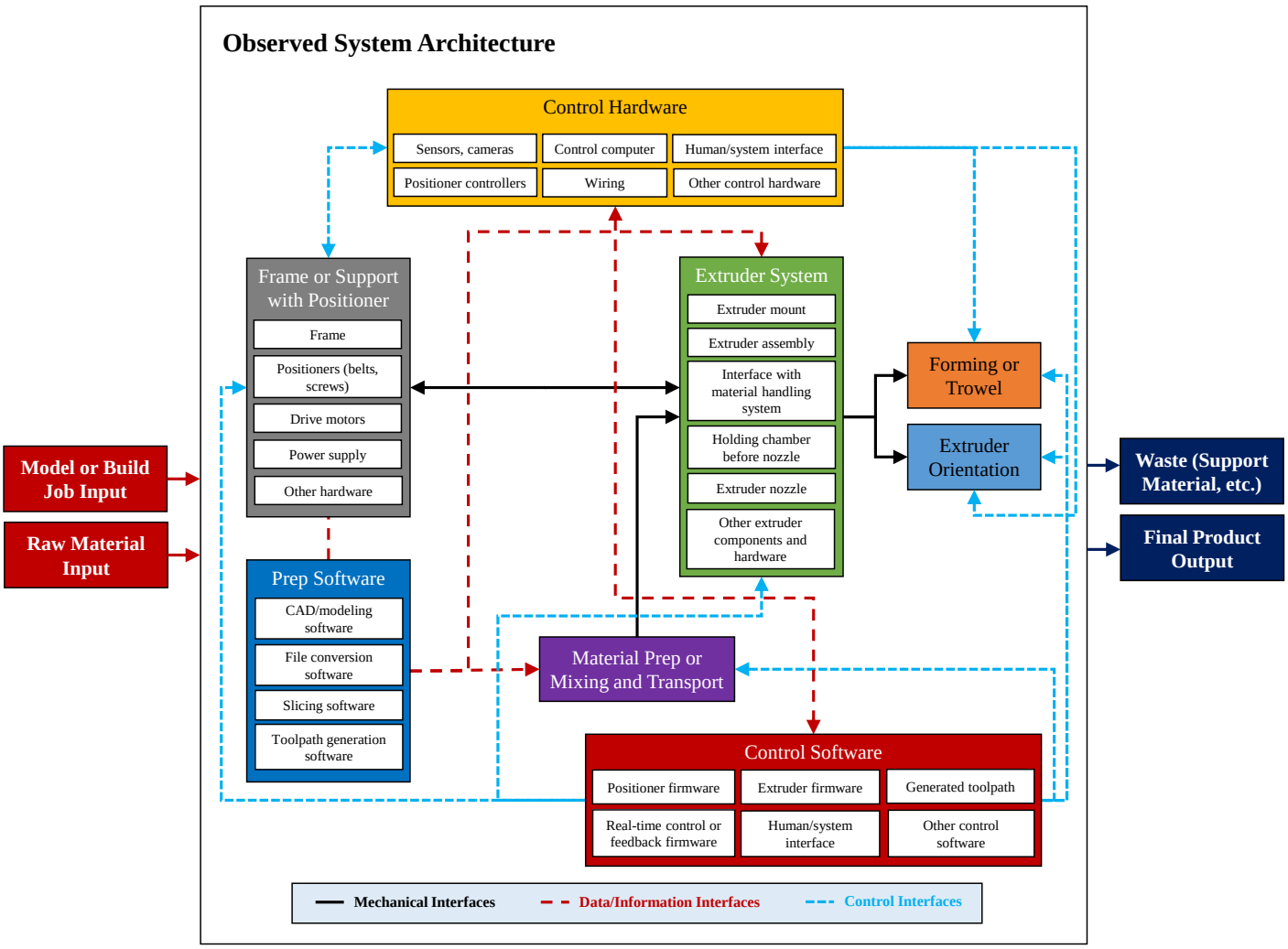

FIGURE 4: *Identified EBACS quasi-general system architecture including interfaces and information flow directions.*

beginning of the study in order to guide and bound the effort; these questions were formulated such that the answers provided vital information toward the development of these preliminary general system architecture, as shown in Figure 4. The collected data was based on the first three research questions:

1. Major system functions
2. Major system and subsystem components
3. Major interfaces within the system (including arrows showing the direction of information exchange - unidirectional or bidirectional)

Each of the reviewed references in Table 2 was examined with these three questions in mind.

The development of this quasi-general system architecture (based on the collected information from existing systems) and eventual refinement of a true general system architecture for EBACSs will prove to be a powerful tool for future implementation. A general perspective on the design and construction of these additive manufacturing systems will provide agility and efficiency for their implementation for the increasingly complex design problems that AM-based construction can be used to address. The architecture presented in this study was meant to service as a starting point, as it is incomplete due to some uncertainties in the collected data and unresolved development issues with EBACSs. In particular, further consideration of the control system hardware, the forming/trowel method, and the design of the material pumping system are needed for a final system architecture. Fortunately, several of the system components appear to be very refined by this point and so are unlikely to change in a major way in the coming years; examples include the positioner design, the control software, and the preparation/pre-processing software. However, further refinements of the software components of the system should be done using ROS and other open-source methods which do not rely on old or proprietary code. In the future, published papers related to EBACSs should include details (at least as supplemental materials) of any new developments, hardware, or software used so the work can be used to drive a general system architecture. Any systems that rely on

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purchased machine packages (for example, Ref [30, 34]) should be characterized as much as possible. System characterization of the common commercially-available systems will be the focus of future work in this area.

Design parameters for EBACSs correspond well to the essential system components and interfaces. While this needs more in-depth study, some of the major design parameters which should be explored with these systems include:

1. **Degree of integration**: The design of the high-level system could range from modular to fully integrated after completion. A more integrated system would be more efficient and reliable for its particular application, but a modular system would provide more flexibility and maintainability and likely would cost less to set up.
2. **Elimination of interfaces**: Related to the degree of integration, eliminating interfaces can increase the reliability of the system.
3. **Extruder design**: The standard extruder system on the reviewed EBACSs appears to be very refined at this point, but further design and development of it could open the door to a wider variety of new materials and possible eliminate the need for the forming/trowel subsystem.
4. **Control hardware**: As discussed in previous sections, the design of the control hardware is not a topic that has been explored much in the reviewed literature. Therefore, many opportunities for improvement and moving toward open-source options surely exist.
5. **Forming/trowel design**: While the designs used in the reviewed literature appeared to be functional, their actual design and control were rarely discussed. An interesting opportunity for future work could be in the design of a forming system which is completely independent of the extruder and can be optimized more easily.
6. **Process monitoring**: Many design opportunities exist for process monitoring, especially solutions that provide a feedback loop to the system and allow detection of defects and machine problems.
7. **Dynamic system optimization**: In any mechanical positioner system, there will be a threat of problems from mechanical vibration, friction, and wear of the system components. The presence of vibration also limits the size of EBACSs and their printing speed. Finding an effective way to predict and deal with the vibration will allow the design of larger and faster printing systems.
8. **Improved autonomy**: While these system are in theory mostly autonomous once the model is fed in, the reviewed literature showed that a significant amount of direct human interaction is still needed for most of the existing EBACS designs. Improvements in the software and in dynamic system optimization should make the system easier to monitor and control automatically, which will make autonomy less risky for these systems.
9. **Human-system interface**: While the eventual goal should be to make these systems fully autonomous, during development and refinement (and for many practical applications), direct human involvement (or at least supervision) is needed. An effective and easy-to-understand interface is needed which can be used while in fully autonomous cases or with a human in the loop.
10. **Autonomous-Manual mode switch**: A very useful design feature that can be added to these systems would be a way for the system to be changed from autonomous to a manual/operator mode depending on the work conditions and potential encountered problems.
11. **Safety and security**: None of the reviewed systems really addressed the need for safety and security systems for EBACS. This is an area of vital importance, not only to protect intellectual property and ensure operator safety, but also because EBACS pose a much larger safety threat to bystanders and the general public than what would be encountered with normal 3-D printers.

6 CONCLUSIONS AND FUTURE WORK

The purpose of this study was to collect a representative sampling of the major literature on extrusion-based autonomous construction systems and use this data to extract a quasi-general system architecture and ideas for design parameters. This is the first step to refining a true system architecture which will be used to design and implement these systems for complex jobs in the future. This will provide a top-down modeling and analysis tool which will be able to guide the assembly, integration, and applications of these machines in the real world. The resulting data showed a number of major areas which need further consideration and development, while proving an incomplete but functional architecture which can be used now and is independent of the system type (assuming it is one of the several different types reviewed). Future work will focus on doing a comprehensive state-of-the-art review of this topic, a more in-depth discussion of design parameters and opportunities, and on filling the identified gaps in order to better refine a general EBACS architecture.

DATA AVAILABILITY

All the collected data from this study is available upon request to the corresponding author.

CONFLICTS OF INTEREST

No external funding was used to perform the work described in this study. Opinions and conclusions presented in this work are solely those of the authors. AEP was on the organizing committee for this conference; he had no role in the evaluation or acceptance of this paper.

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